

Assignment 4

Total: 5 marks

This assignment must be hand-written. **Show ALL steps in ALL questions.**
Read the questions carefully.

You are designing a digital volume control system for a speaker that adjusts the volume between four levels: Mute (0), Low (1), Medium (2), and High (3). However, there is a restriction: Transitions directly between Mute (0) and High (3), or High (3) and Mute (0) are not permitted. The system should track the current volume level using a 2-bit up/down counter.

Now, design the 2-bit up/down counter using T flip flop(s).

Note: The circuit is to be designed by treating the unused states (if there is any) as Don't Care conditions.